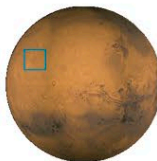


THARSIS REGION

Olympus Mons



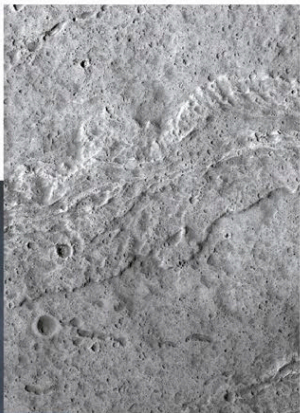
TYPE Shield volcano
AGE 30 million years
DIAMETER 403 miles (648 km)

Olympus Mons is unquestionably the largest volcano in the solar system. Its height, of about 15 miles (24 km), makes it the tallest, and its volume is over 50 times that of any shield volcano

on Earth. Olympus is one of the giant shield volcanoes of the Tharsis region, which is home to the greatest number of volcanoes on Mars, including the planet's youngest. Volcanoes evolve over long periods of time and can be inactive for hundreds of millions of years. Olympus Mons is considered the youngest of the shield volcanoes. The summit has a complex caldera.

Different areas of its floor are associated with different periods of activity. The largest central area, which is marked by ring-shaped faults, is more recent, at 140 million years old.

The caldera is surrounded by a surface of wide terraces formed by lava flows, crossed by thinner flows. These are encircled by a huge scarp, up to 3.7 miles (6 km) high. Vast plains, termed aureoles, extend from the north and west of the summit like petals from a flower. These regions of gigantic ridges and blocks, whose origins remain unexplained, extend outward for up to 600 miles (1,000 km).



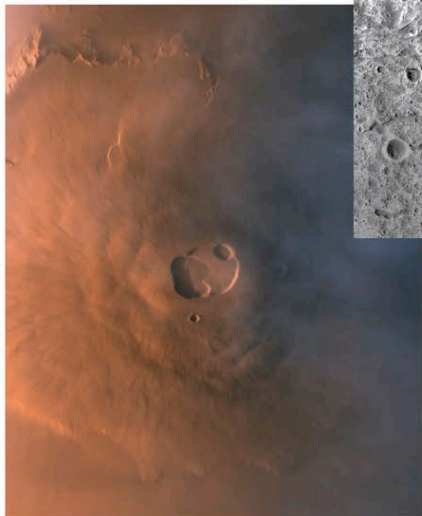
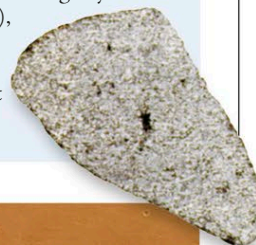
LAVA FLOWS

These lava flows and a collapsed lava tube (top left) on the southwest flank have been peppered by tiny impact craters.

EXPLORING SPACE

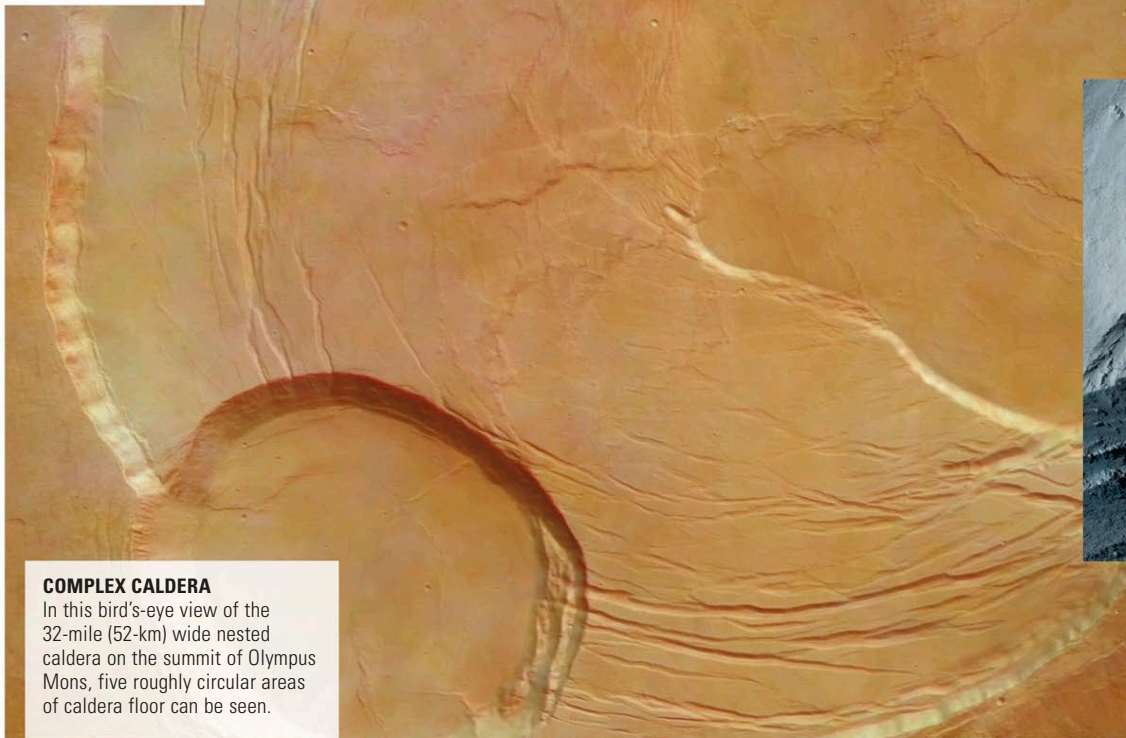
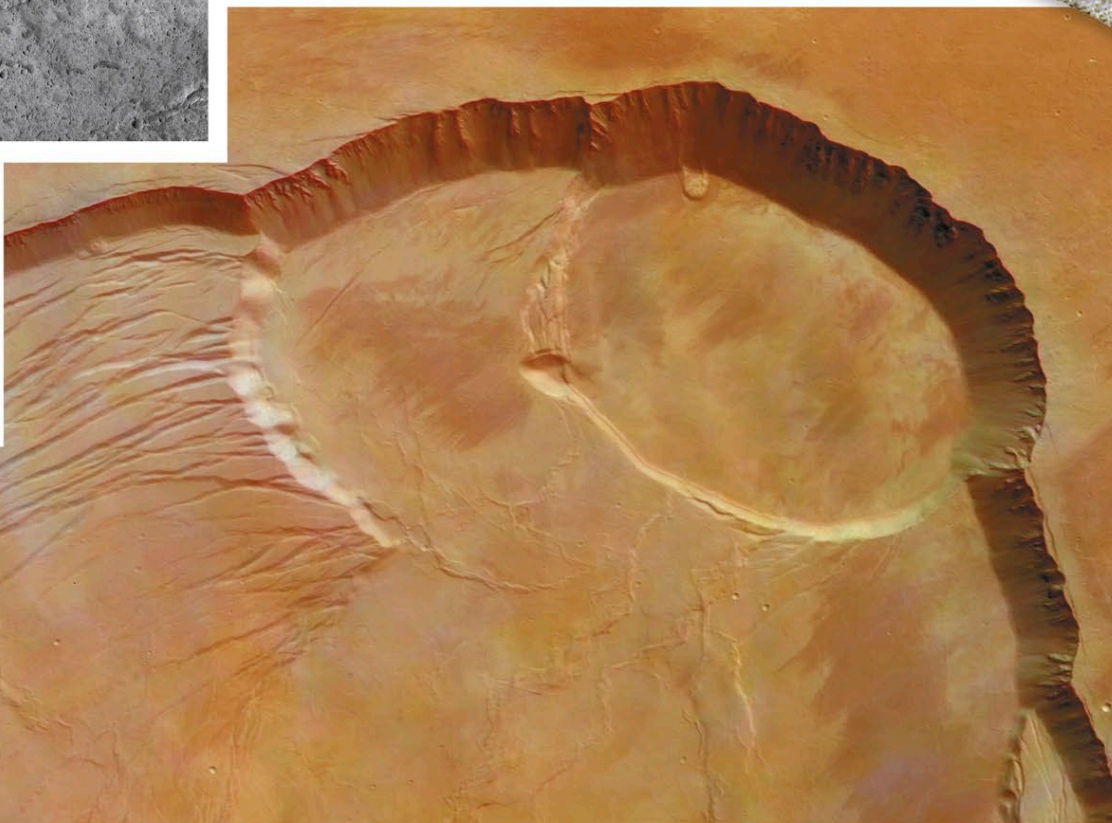
MARTIAN METEORITES

Solidified basaltic lava covers the Tharsis region. Pieces of lava that flowed on the Martian surface as recently as 180 million years ago are now on Earth. Impactors hit Mars and ejected them, and after journeys lasting millions of years, they fell to Earth as meteorites. They include the Shergotty meteorite (right), which landed in Shergahti, India, on August 25, 1865.



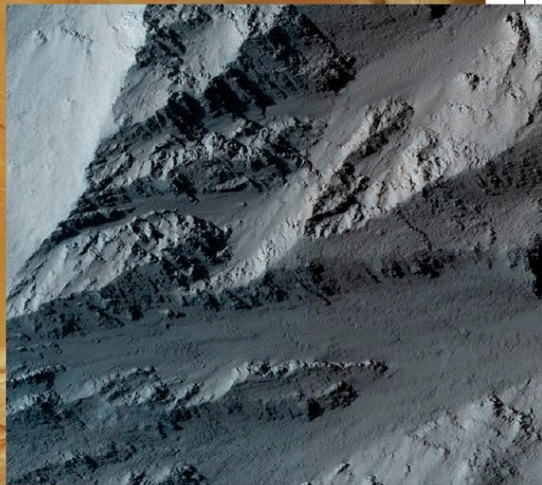
MIGHTY OLYMPUS

This massive volcano is named after the mountaintop home of the gods and goddesses of Greek mythology. Broad lava-flow terraces surround the caldera at the volcano's summit.



COMPLEX CALDERA

In this bird's-eye view of the 32-mile (52-km) wide nested caldera on the summit of Olympus Mons, five roughly circular areas of caldera floor can be seen.



LANDSLIDE CLIFFS

Olympus Mons is bounded on all sides by steep cliffs, thought to have been caused by landslides. This close-up taken by Mars Reconnaissance Orbiter shows an area of the cliffs about 0.6 miles (1 km) wide on the northern side of the volcano.